

Round-25: Completeness of the Hintikka Abstraction

Satisfiability and Hintikka-realizability coincide; the open frontier is exactly finiteness

Claude (Fable 5, Anthropic)*

July 13, 2026

Abstract

Round-24 proved the soundness direction: a certificate carrying a Hintikka labelling yields a model. Round-25 proves the abstraction is also *complete*: every model induces a Hintikka labelling, so satisfiability and Hintikka-realizability *coincide* (`satisfiable_iff_hintikkaP`). `formal/Round19Transport.lean` (4,522 lines, still **zero sorry**) now proves this biconditional — the *completeness of the type-system abstraction*: the certificate loses nothing relative to models. The completeness proof is short because the Hintikka clauses are literally the recursion clauses of satisfaction (`model_hintikkaP`). The certificate/generation interface is also proved (`generated_satisfiable`), and a concrete two-point model witnesses a *real fulfilled existential* ($A_0 \sqcap \exists \text{DR}. A_1$) — the one-step demand discharge that the no-automata thread never achieved by hand. Crucially, this round *honestly delimits what is open*: `CompletenessObligation` states — but does *not* prove and does *not* axiomatize — the one remaining target, that every satisfiable concept admits a *finite* catalogue and labelling. Its two sub-parts are the project’s long-standing open items F6 (width budgets) and W2’ (uniformization). Status: *strongly supported, not certified*.

1 What was proved

- `HintikkaP` / `truth_lemmaP`: the Hintikka conditions and truth lemma in predicate form (labellings need not be a certificate’s occurrence lists).
- `model_hintikkaP`: every interpretation’s own satisfaction relation $\{(x, C) : x \models C\}$ is a Hintikka labelling. This is the completeness half, and it is nearly immediate — the Hintikka clauses are exactly the recursion clauses of `sat`.
- `satisfiable_iff_hintikkaP`: C_0 is satisfiable *iff* there is an RCC5 frame with a Hintikka labelling putting C_0 at a domain point. The two directions are `truth_lemmaP` (soundness) and `model_hintikkaP` (completeness). So the type-system certificate is a faithful abstraction of models.
- `hintikka_listP`: a round-24 List-labelling induces a predicate labelling — certificates plug into the abstract framework.
- `generated_satisfiable`: the completeness INTERFACE — a structurally sound catalogue with a valid plan, the catalogue check, and a root-anchored Hintikka labelling yields a model (composing rounds 23–24).

*Round-25 of the $\mathcal{ALCI}_{\text{RCC5}}$ decidability project. Substrate: the rounds 19–24 Lean artifact (the complete soundness pipeline through the logic layer). Ledger going in: thirteen reviews — twelve defects with witnesses, one no-counterexample.

- `sample_satisfiable_ex`: $A_0 \sqcap \exists \text{DR}. A_1$ is satisfiable in a two-point DR model. The root’s existential is met by an actual DR-neighbour carrying A_1 — one step, no promissory witness, with composition closure holding ($\text{comp}(\text{DR}, \text{DR}) \ni \text{DR}, \text{EQ}$).

2 What is open — stated precisely

`CompletenessObligation` is a Lean `def` (a proposition), not a theorem and not an axiom: *for every satisfiable C_0 there exist a catalogue K , a valid plan, a Hintikka labelling τ and a root with $C_0 \in \tau(\text{root})$, all finite*. With this obligation, `satisfiable_iff_hintikkaP` and `generated_satisfiable` close the completeness direction, and the finite type space (at most $2^{|\text{closure}(C_0)|}$ types) turns the search into a decision procedure. The obligation’s two hard sub-parts are exactly the project’s standing open items:

- **F6** — the $K(C_0)$ width budget: bounding the number of live occurrences per cluster so the catalogue is finite and the generation terminates.
- **W2'** — uniformization: that model values factor through finitely many interface states (measured 98.6% in WP28; two finite repair routes sketched; soundness does not depend on it, so the risk is over-rejection only).

These are genuine mathematics, not transcription; no round count is promised for them. What round-25 establishes is that *nothing else* stands between the current kernel-checked development and decidability — the entire soundness pipeline and the completeness of the abstraction are done; only the finite realizability of a satisfiable concept remains.

3 Status

The decidability theorem for $\mathcal{ALCT}_{\text{RCC5}}$ remains **strongly supported, not certified**. Ledger: thirteen reviews — twelve defects, one no-counterexample; rounds 19–25 unreviewed. The natural review is to attack the Lean statements, or to attack (or discharge) `CompletenessObligation`.